Lab 4 Report Microprocessor

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1 PT-51 Kit Verification and Peripheral Testing

1.1 Objective

To familiarize with the PT-51 (8051-based) development kit, set up the necessary programming environment (Atmel FLIP or DFU Programmer), and verify the functional integrity of the board's peripherals using pre-compiled hex files (led.hex and pt51_test.hex).

1.2 Observation

- Bootloader Mode: The board successfully entered bootloader mode using the specific sequence (Pressing Boot + Reset, then releasing Reset then Boot). The computer recognized the device as an AT89C5131A
- Software Connectivity: After installing the JRE and FLIP/DFU, the software was able to open the USB port and communicate with the 8051 microcontroller
- LED Toggle Test: Upon flashing led.hex, the on-board LEDs began to toggle at a regular interval, confirming that the CPU is executing code and the pins are functional
- Switches: Responded to inputs indicated by the test program status
- LCD: Displayed test strings or status messages

1.3 Conclusion

The PT-51 kit was successfully configured and verified. The installation of the FLIP software and Java Runtime Environment (JRE) allowed for seamless uploading of hex files. Since all peripheral tests in pt51_test.hex passed without error, the hardware is confirmed to be in good working condition and ready for further assembly language or C programming tasks.

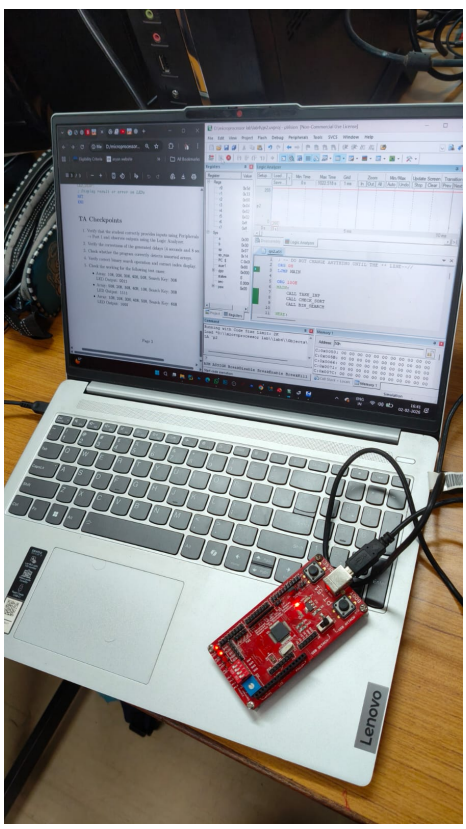
2 Array Sorting Verification

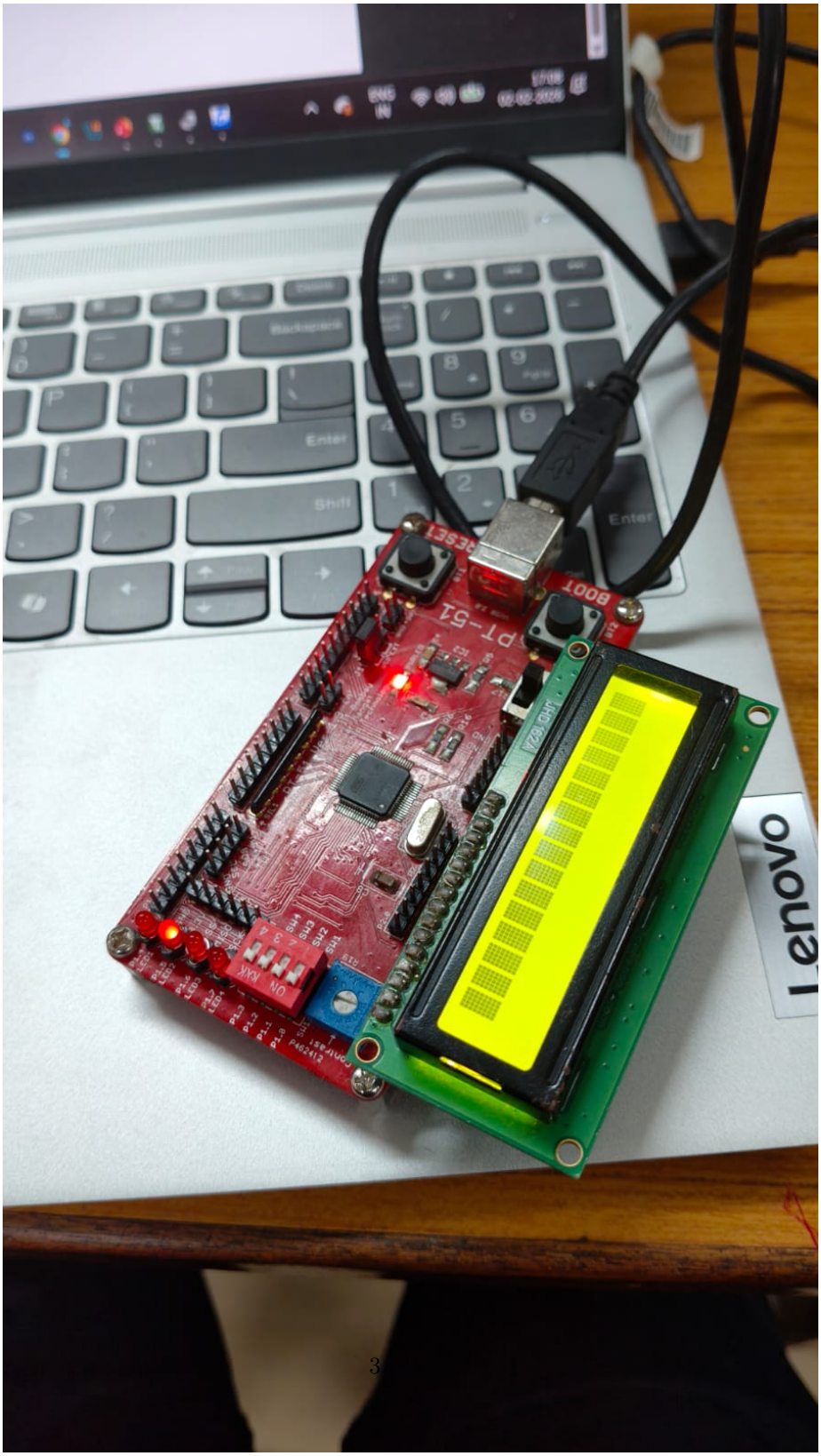
2.1 Objective

To implement an 8051 Assembly program that:

- Reads six 8-bit inputs via DIP switches using a nibble-by-nibble method with precise timing delays
- Validates if the first five inputs (the array) are stored in ascending order
- Executes a Binary Search algorithm to find a 6th input-the key within the array
- Communicates system status (Not Sorted, Not Found, Index Found) via on-board LEDs

2.2 Results




RESET
BOOT
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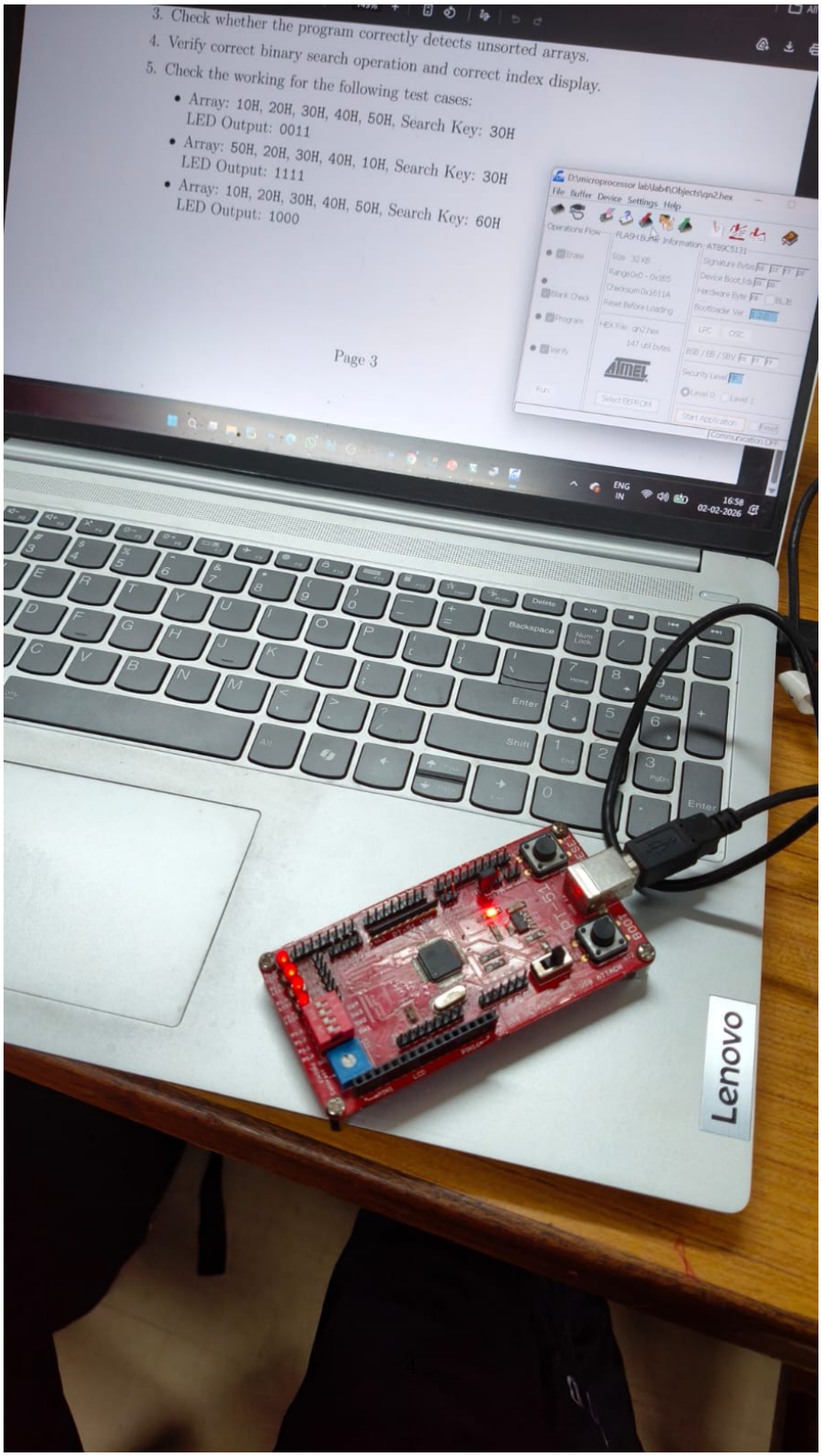
3. Check whether the program correctly detects unsorted arrays.
4. Verify correct binary search operation and correct index display.
5. Check the working for the following test cases:
• Array: 10H, 20H, 30H, 40H, 50H, Search Key: 30H
LED Output: 0011
• Array: 50H, 20H, 30H, 40H, 10H, Search Key: 30H
LED Output: 1111
• Array: 10H, 20H, 30H, 40H, 50H, Search Key: 60H
LED Output: 1000
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2.3 Observations

- Input Mechanism: The 8-second delay provided sufficient time to toggle P1.0–P1.3 for the upper nibble, followed by the lower nibble. The LEDs correctly incremented in binary, acting as a progress tracker for the 12 total nibble entries (6 full bytes)
- Sorting Logic: When the array was entered out of sequence (e.g., 50H followed by 20H), the program immediately identified the "Unsorted" state and halted with the correct error code
- Search Logic: For sorted arrays, the binary search correctly split the range and identified the target index

2.4 Conclusion

- Time Management: Using nested loops to scale a 1ms delay into 4s and 8s intervals is essential for humam-interfaced inputs
- I/O Constraints: The "nibble-shift" method (using ANL, ORL, and SWAP instructions) effectively bypassed the hardware limitation of having only 4 physical switches for 8-bit data
- Final Result: The program met all TA checkpoints, correctly identifying sorted states and accurately retrieving element positions